Name : Priya Sharma

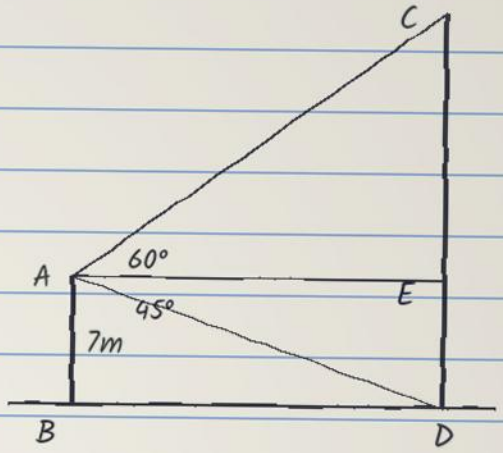
Class : X - B Roll No. 05

Section D

Ans 18.

Let AB = 7 m be the building , CD the tower ,

BD = AE = x



In ΔABD , $\tan 45° = \frac{AB}{BD}$

$1 = 7/x \Rightarrow x = 7$ m

In ΔAEC , $\tan 60° = \frac{CE}{AE}$

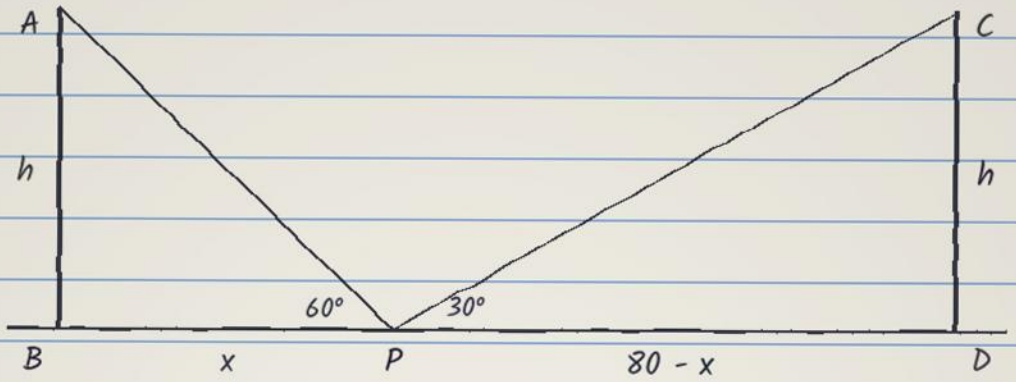
$\sqrt{3} = CE/7 \Rightarrow CE = 7\sqrt{3}$ m

$CD = CE + ED = 7\sqrt{3} + 7 = 7(\sqrt{3} + 1)$

$= 7 \times 2.732 = 19.12$ m

$\therefore$ Height of tower = 19.12 m

Ans 19.



Let $BP = x$, $PD = 80 - x$, $AB = CD = h$

In ΔABP : $\tan 60° = h/x \Rightarrow h = \sqrt{3}\,x$ ...(i)

In ΔCDP : $\tan 30° = \dfrac{h}{80 - x} \Rightarrow h = \dfrac{80 - x}{\sqrt{3}}$ ...(ii)

(i) = (ii) : $\sqrt{3}\,x = (80 - x)/\sqrt{3}$

$3x = 80 - x \Rightarrow 4x = 80 \Rightarrow x = 20$ m

$h = 20\sqrt{3} = 20 \times 1.732 = 34.64$ m

Height of each pole = 34.64 m

Distances : $BP = 20$ m , $PD = 80 - 20 = 60$ m

Ans 17. LHS $= \dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta}$

$= \dfrac{\tan\theta}{1-1/\tan\theta} + \dfrac{1/\tan\theta}{1-\tan\theta}$

$= \dfrac{\tan^2\theta}{\tan\theta - 1} - \dfrac{1}{\tan\theta\,(\tan\theta - 1)}$

$= \dfrac{\tan^3\theta - 1}{\tan\theta\,(\tan\theta - 1)}$

(contd. on next page)

Q17 contd.

$$= \frac{(\tan\theta - 1)(\tan^2\theta + \tan\theta + 1)}{\tan\theta\,(\tan\theta - 1)} [a^3 - b^3]$$

$$= \frac{\tan^2\theta + \tan\theta + 1}{\tan\theta} = \tan\theta + 1 + \cot\theta$$

$$= 1 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 1 + \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\,\cos\theta}$$

$$= 1 + \frac{1}{\sin\theta\,\cos\theta} = 1 + \sec\theta\,\operatorname{cosec}\theta = RHS$$

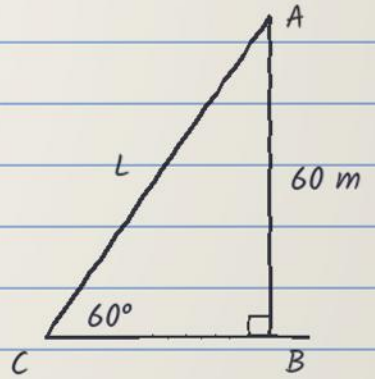
Section C

C-4.

$AB = 60$ m (height), $AC = L$ (string)

$$\sin 60^\circ = \frac{AB}{AC} = \frac{60}{L}$$

$$\frac{\sqrt{3}}{2} = \frac{60}{L} \Rightarrow L = \frac{120}{\sqrt{3}} = 40\sqrt{3}\ m$$



C-3. $LHS = \frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A}$

$= \frac{\cos^2 A + (1+\sin A)^2}{\cos A\,(1+\sin A)} = \frac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{\cos A\,(1+\sin A)}$

$= \frac{2 + 2\sin A}{\cos A\,(1+\sin A)} = \frac{2\,(1+\sin A)}{\cos A\,(1+\sin A)} = 2\sec A = RHS$

Q14. $\sec\theta = 13/12 \Rightarrow \cos\theta = 12/13$

$\sin\theta = \sqrt{(1 - 144/169)} = 5/13$

$\tan\theta = 5/12$, $\cot\theta = 12/5$, $\operatorname{cosec}\theta = 13/5$

Q13.

~~$(\sec A - \tan A)^2 = \sec^2 A - \tan^2 A = 1$~~

~~$= 1 \neq RHS$~~

$LHS = \left(\frac{1}{\cos A} - \frac{\sin A}{\cos A}\right)^2 = \frac{(1-\sin A)^2}{\cos^2 A}$

$= \frac{(1-\sin A)^2}{1-\sin^2 A} = \frac{(1-\sin A)^2}{(1-\sin A)(1+\sin A)} = \frac{1-\sin A}{1+\sin A}$

Sec B

Q12. $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$= \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$

$= 2(\sin^2\theta + \cos^2\theta) = 2 \times 1 = 2 = RHS$

Q11. $\sin(A - B) = 1/2 \Rightarrow A - B = 30°$

$\cos(A + B) = 1/2 \Rightarrow A + B = 30°$

$2A = 60° \Rightarrow A = 30°, B = 0°$

Q10. $2(1)^2 + (\sqrt{3}/2)^2 - (\sqrt{3}/2)^2 = 2 + 3/4 - 3/4 = 2$

Q9. $\cot A = 4/3$, $\tan A = 3/4$, $\tan^2 A = 9/16$

$$\frac{1 - 9/16}{1 + 9/16} = \frac{7/16}{25/16} = 7/25$$

Section A

1 - (a) 2 - (c) 3 - (a) 4 - (b)

5 - (b) 6 - (b) 7 - (c) 8 - (a)